Tools used

Relative strength index:

$$Rsi = 100 - \left(\frac{100}{1 + \frac{\text{Average gains}_n}{\text{Average losses}_n}} \right)$$

```
def calculate_rsi(series, period=14):
    delta = series.diff()
    gain = (delta.where(delta > 0, 0)).rolling(window=period).mean()
    loss = (-delta.where(delta < 0, 0)).rolling(window=period).mean()
    rs = gain / loss
    return 100 - (100 / (1 + rs))
```

Exponential moving average:

$$Ema_t = \left(\text{Price}_t \times \frac{2}{n+1} \right) + Ema_{t-1} \times \left(1 - \frac{2}{n+1} \right)$$

```
df['ema200'] = df['Close'].ewm(span=200, adjust=False).mean()
```

Logic

Position management:

Entry occurs only if the momentum (Rsi) is inconsistent with the trend (Ema), suggesting an imminent correction.

```
if position == 0:
    if rsi < 40 and price > ema200:
        position = 1
        entry_price = price
        balance -= balance * fee # Deduzione commissione 0.1%
        trades += 1
    elif rsi > 60 and price < ema200:
        position = -1
        entry_price = price
        balance -= balance * fee
        trades += 1
```

Exits and risk management:

The trade is closed for three reasons: reaching the target (Take profit), protecting against risk (Stop loss) or a reversal of momentum.

```
elif position == 1:
    pnl = (price - entry_price) / entry_price
    if rsi > 55 or pnl >= tp or pnl <= -sl:
```

```
        balance += balance * pnl
        balance -= balance * fee
        position = 0
```

The 60 day dataset was divided into 24 hour windows to observe daily performance.

```
window_size = 24
num_windows = len(df) // window_size

for i in range(num_windows):
    df_slice = df.iloc[start_idx:end_idx].reset_index(drop=True)
    final_val, trades = run_single_backtest(df_slice)
    profit = final_val - 1000
```

Results

Metric	Value
Initial capital	1000.00 Usdt
Total profit	+ 2.25 Usdt
Profitable days	4 / 60 (6.7 %)
Average daily return	0.00 %
Effectiveness	Low frequency, low risk